

Your name and entry number:

1. You may have noticed a small paratha shop just outside the IIT Delhi metro Station. The person who runs the shop is also the one who cooks the parathas and collects money from customers. These parathas are sold for ₹30 per piece (if I remember correctly!). Of course, no bills are issued.

A restaurant in Hauz Khas village sells similar (in taste and form, at least) parathas for ₹200 per piece. The entrepreneur who runs the restaurant has taken the building on lease from a property developer. Cooking is done by one of the staff members.

As you can imagine, when one paratha is produced and sold, what enters the GDP calculation is ₹30 in shop 1 and ₹200 in shop 2. We know that production leads to the generation of income.

Remember the formula:  $GDP = (PX - aPX) = wbX + \Pi PX + T_x$

Here X can be thought of as one (one paratha) and P as ₹30 in case 1 and ₹200 in case 2.

1a. Who all receives income in the case of shop (1), and what is the nature of that income (labour, capital, anything else??)

1a. Self-employed person receiving mixed income. 1 mark

1b. What about shop 2? Who do you think receives the biggest share of income? Any comments?

1b. In the second case, there is labour income ( $wbX$ ) for the worker. Then at least two types of capital incomes (together, it is  $\Pi PX$ ) – rent for the property developer and profit for the entrepreneur. The government receives indirect taxes in case 2, through GST ( $T_x$ ).

2 marks

In both the above cases, the suppliers of inputs (wheat, etc.) (cost of inputs:  $aPX$ ) also receive incomes.

In the second case, it is likely that rent is one of the largest components of income (but that income reaches persons who do not take an active part in the production process).

In the first case, the self-employed person is making very little income per each paratha sold. She or he must work really long hours to earn their livelihood

2. The Table below shows the components of the money supply in India (in trillions of Rupees).  
 (i) What was the value of all deposits of the (commercial) banks in 2022-23? (ii) In 2022-23, if the cash in banks was Rs.1 trillion, what was the value of the commercial banks' deposits with the Reserve Bank of India? What were the values of the money multiplier for 2016-17 and 2022-23?

|         | M0 or Reserve Money | M3 or Broad Money | Currency with the public |
|---------|---------------------|-------------------|--------------------------|
| 2016-17 | 19                  | 128               | 13                       |
| 2022-23 | 44                  | 223               | 33                       |

In 2022-23, Money supply (M3 or M) is ₹223 trillion.

$M = \text{Currency with the public} + \text{Deposits}$ ; **Deposits = ₹190 trillion** 1 mark

Reserve Money or M0 = Currency with the public + Cash in bank+ Commercial banks' deposits (part of reserves) with the RBI = ₹44 trillion

$44 = 33 + 1 + \text{Deposits with RBI}$

**Commercial banks' deposits with the RBI = ₹10 trillion.** 1 mark

Money multiplier:  $M3/M0$ . In 2016-17 =  $128/19$

In 2022-23 =  $223/44 =$  1 mark

3. Which of the following countries is a net recipient of foreign savings? China, India, U.S.?  
 Are their current accounts in surplus or in deficit?

Both India and the U.S. have current account deficits.

**A country with a current account deficit is a net recipient of foreign savings.**

1 mark

China, which has a current account surplus, is a net donor of savings to the rest of the world.

4. Remember the relation for the assets and liabilities of businesses:

$$qPK = L_B + eZ_B^* + P_V + Q_B$$

A few of the big businesses have received large loans from banks, and they have also been raising capital from the stock market. The share prices of these companies were rising for several months. However, some sudden economic developments have led to a big fall in share prices. How does that affect the above-referred balance sheet of businesses? Will this event (fall in share prices) have any impact on the balance sheets of households and banks? 2 marks

When  $P_v$  (price of each share) rises and later falls, the changes will be reflected in the value of  $q$  (Tobin's  $q$  which is the ratio of market value to replacement cost).

During a boom period (when share prices were rising), the businesses are able to raise large volumes of financial capital. The boom in the stock market encourage banks also to provide more and bigger loans to the businesses.

During a slump, however, a good part of the resources raised through the stock market could be wiped off (fall in both  $P_v$  and  $V$ ). That will have implications for banks and households.

The assets and liabilities of banks, businesses, and households

|            | Liabilities              | Assets          |
|------------|--------------------------|-----------------|
| Business   | $L_B + P_v V + \Omega_B$ | $qPK$           |
| Banks      | $M$ (currency+deposits)  | $L (L_B + L_G)$ |
| Households | $\Omega_H$               | $M + P_v V$     |

With the crash in the stock market, there will be concerns whether businesses will be able to repay the loans they have taken from banks. The balance sheets of banks become unstable too. If some of the loans are not going to be repaid, will banks be able to maintain the value of the deposits they hold?

These bank deposits and currencies are the assets or accumulated savings of the households. If the households fear that they lost their savings, they will reduce purchases, triggering an economic recession.

5. Write external debt as % of GDP using the algebraic framework we have learnt?

$$(eZG^* + eZB^*)/PX$$

1 mark